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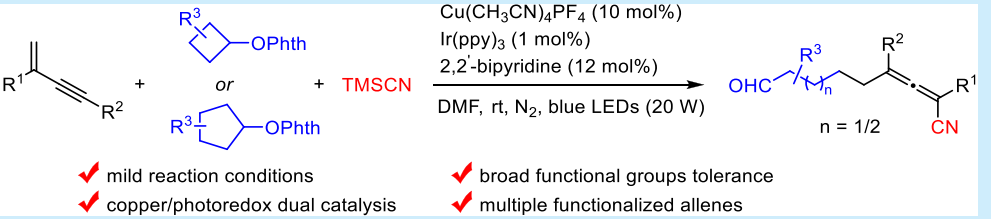
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Three-Component Synthesis of Multiple Functionalized Allenes via Copper/Photoredox Dual Catalyzed 1,4-Alkylcyanation of 1,3-Enynes

Yue Pu,[‖] Shumin Ding,[‖] Haixia Zhao, Qiaoli Xue, Heng Zhang, Xiaotian Xie, Yongjia Shang,* and Jian Wang*

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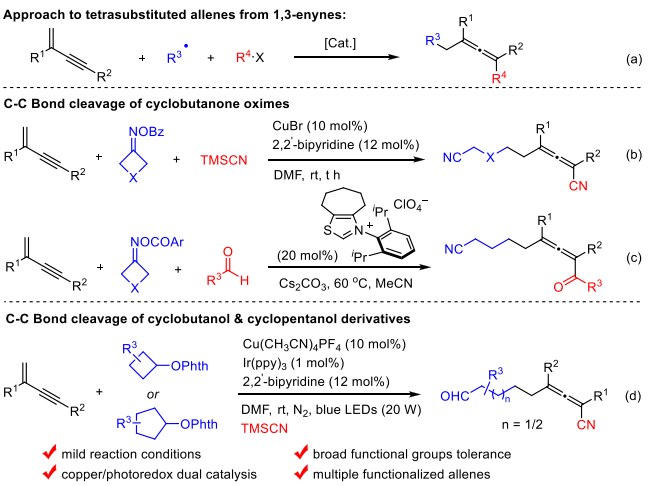


ABSTRACT: Efficient access to multiple functionalized allenes via a three component 1,4-alkylcyanation of enynes with cyclic alcohol derivatives in the presence of trimethylsilyl cyanide (TMSCN) under copper/photoredox dual catalysis has been developed. Both easily transformable aldehyde and cyano groups were introduced to tetra-substituted allenes through light-induced C−C bond cleavage of cyclic butanol and pentanol derivatives. The reactions proceeded smoothly under mild conditions with broad functional groups tolerance.

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Allenes represent an important class of molecules, which are ubiquitous in natural products and pharmaceuticals as well as functional materials.[1] In addition, they play crucial roles in organic synthesis and drug design.[2] Thus, continuous efforts were devoted to the assembly of functionalized allenes, and several strategies such as sigmatropic rearrangement, β-eliminations, and nucleophilic substitutions have been successfully developed.[3] However, the aforementioned traditional methods mainly relied on prefunctionalization of starting materials. Conjugated 1,3-enynes bearing multiple reactive sites have been established as important synthons in organic synthesis.[4] Recently, transition-metal-catalyzed radical-involving difunctionalization of 1,3-enynes has emerged as an appealing and powerful tool for the construction of multisubstituted allenes (Scheme 1a).[5] Although a number of elegant synthetic methods have been reported by Ma,[6] Liu,[7] Bao,[8] Wu,[9] Li,[10] and other groups,[11] 1,4-dicarbofunctionalization of enynes to afford tetrasubstituted allenes has been less explored.[12] For instance, in 2019, trasition-metal-catalyzed 1,4-alkylarylation of enyes using boronic acids as nucleophiles was reported by Wang[13] and Bao,[14] respectively, offering rapid access to tetra-substituted aryl allenes. In 2022, Kong and coworkers developed an efficient synthesis of multifunctionalized CF_3-allenes via nickel/photoredox catalysis.[15] Very recently, a cross-electrophile 1,4-alkylarylation of 1,3-enynes using alkyl and aryl bromides was reported by Deng, Chen, and coworkers.[16]

Scheme 1. Catalytic 1,4-Dicarbofunctionalization of Enynes through C−C Bond Cleavage



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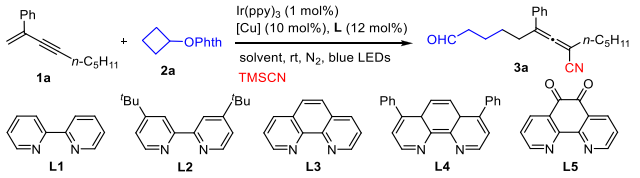
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Strained ring derivatives such as cyclobutanone oxime esters and substituted cyclopropanols have been well established as precursors of functionalized primary radicals via C−C bond cleavage.[17] In addition, alkylcyanation of unsaturated groups afforded efficient approaches to functional molecules with multiple C−C bond formations.[18] In 2021, the Ma group reported a pioneering copper-catalyzed three component 1,4-alkylcyanation of enynes using cyclobutanone oxime esters in the presence of trimethylsilyl cyanide (TMSCN) or trimethyl-(trifluoromethyl)silane ($TMSCF_3$), providing facile access to 1,7-double-functionalized allenes (Scheme 1b).[6a] In the same year, a novel N-heterocyclic carbene (NHC)-catalyzed radical 1,4-alkylcarbonylation of 1,3-enynes was developed by Du and co-workers (Scheme 1c).[19a] However, these reports were mostly limited to the construction of 1,7-double-functionalized allenes. The development of novel 1,4-dicarbofunctionalization methods of 1,3-enynes for synthesizing diverse multiple-functionalized allenes is in continuous demand. Herein, we have developed an efficient approach to access 1,7- and 1,8-double-functionalized allenes via a three component 1,4-alkylcyanation of enynes under copper/photoredox dual catalysis (Scheme 1d). Both cyclic butanol and pentanol derivatives were suitable in the reaction, serving as efficient aldehyde-containing primary radical precursors via a C−C bond cleavage.[20] By using this method, easily transformable aldehyde and cyano groups were introduced to tetra-substituted allenes under mild reaction conditions in the presence of TMSCN.

The study was initiated with non-1-en-3-yn-2-ylbenzene (**1a**), 2-cyclobutoxyisoindoline-1,3-dione (**2a**), and TMSCN as model substrates using $Ir(ppy)_3$ as a photocatalyst with 20 W blue LEDs as the light source under nitrogen atmosphere at room temperature (Table 1). To our delight, when the reaction was carried out with $Cu(MeCN)_4PF_6$ as the cocatalyst and 2,2′-bipyridine (L1) as the ligand in dichloromethane, the desired product **3a** was generated in 50% yield (entry 1). Screening of solvents (entries 2−6) suggested that polar solvents such as MeCN and DMF were more effective, and product **3a** was obtained in 93% and 98% yields, respectively. The reaction was almost of equal efficiency when alkyl substituted ligand L2 was used with DMF as the ideal choice of solvent (entry 7). The reaction yield of **3a** decreased to 43−81% with L3−L5 as ligands (entries 8−10). Next, other copper sources such as $Cu(MeCN)_4BF_4$, $Cu(OAc)_2$, CuI, CuCl, and CuBr were tested, leading to a sharp decrease of the reaction yield (entries 11−15). The reaction was totally suppressed under the dark (entry 16).

With the optimized conditions in hand, we then moved our attention to the substrate scope of 1,3-enynes (**1**) with 2-cyclobutoxyisoindoline-1,3-dione (**2a**) as the reaction partner in the presence of TMSCN (Scheme 2). Substitution of the alkyne site (R^1) was first investigated. Alkyl groups such as *n*-pentyl, *t*-butyl, and *i*-butyl substituted products **3a**−**3c** were generated in moderate to excellent yields. Compound **3b** was obtained in lower yield, suggesting that the reactivity of the allenyl radical was sensitive to steric hindrance under the applied conditions. Cyclic alkyl groups including cyclopropyl allenes **3d**−**3f** were successfully achieved in excellent yields. Functionalized aryl-substituted enynes participated in the reaction smoothly, offering **3h**−**3k** in moderate to good yields. It is worthy to point out that aryl bromide remained stable under the standard conditions. Satisfyingly, functional groups such as Cl, Br, OAc, NHBoc, CN, and protected amines were

Table 1. Optimization of Reaction Conditions[a]



entry	catalyst	ligand	solvent	yield (%)
1	$Cu(MeCN)_4PF_6$	L1	DCM	50
2	$Cu(MeCN)_4PF_6$	L1	EA	0
3	$Cu(MeCN)_4PF_6$	L1	$PhCF_3$	trace
4	$Cu(MeCN)_4PF_6$	L1	dioxane	0
5	$Cu(MeCN)_4PF_6$	L1	MeCN	93
6	$Cu(MeCN)_4PF_6$	L1	DMF	98
7	$Cu(MeCN)_4PF_6$	L2	DMF	91
8	$Cu(MeCN)_4PF_6$	L3	DMF	43
9	$Cu(MeCN)_4PF_6$	L4	DMF	40
10	$Cu(MeCN)_4PF_6$	L5	DMF	81
11	$Cu(MeCN)_4BF_4$	L1	DMF	72
12	$Cu(OAc)_2$	L1	DMF	53
13	CuI	L1	DMF	61
14	CuCl	L1	DMF	13
15	CuBr	L1	DMF	20
16[b]	$Cu(MeCN)_4PF_6$	L1	DMF	0

[a]Reaction conditions: **1a** (0.1 mmol), **2a** (0.15 mmol), TMSCN (0.25 mmol), $Ir(ppy)_3$ (1 mol %), [Cu] (10 mol %), L (12 mol %), solvent (2.0 mL), 20 W blue LEDs, N_2, 24 h. Yield of isolated products. [b]Under dark.

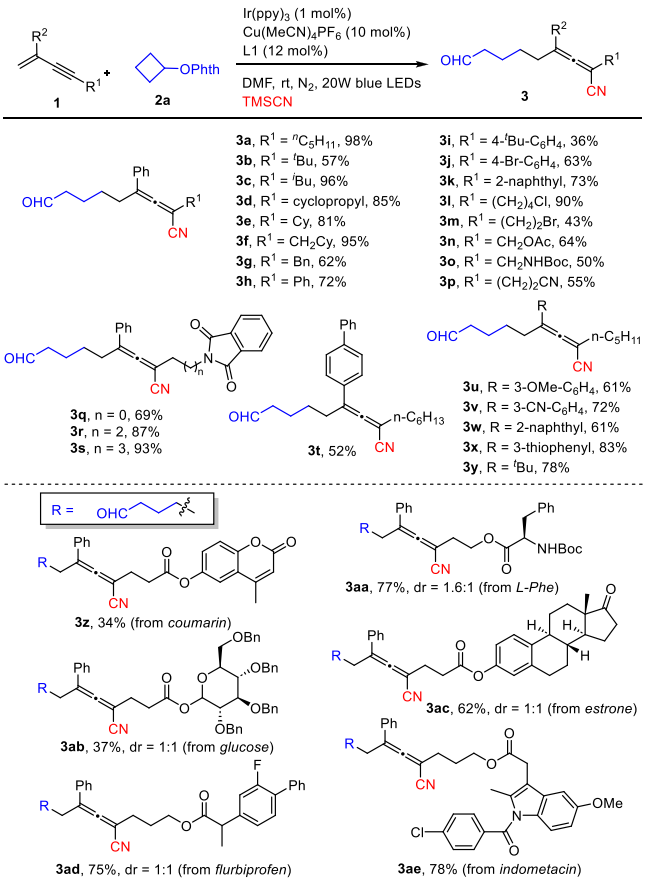
well tolerated, affording the corresponding products **3l**−**3s** in moderate to excellent yields. Next, the R^2 group of 1,3-enynes was studied. Substrates bearing both electron-donating (OMe) and -withdrawing (CN) groups of the phenyl ring proceeded the reaction efficiently, delivering products **3u** and **3v** in 61% and 72% yields, respectively. Naphthyl-, heteroaryl-, and alkyl-substituted products **3w**, **3x**, and **3y** were also given in good yields. To further demonstrate the utility of this reaction, we tested the three component allenation reaction with 1,3-enynes derived from natural products (coumarin, L-phenylalanine, glucose, and estrone) and drug molecules (flurbiprofen and indometacin). The products **3z**−**3ae** were uniformly obtained in moderate to good yields.

Next, the scope of cyclic alcohol derivatives **2** was explored (Scheme 3). Functional group substituted substrate **2b** participated in the reaction efficiently, offering access to the corresponding products **4a** and **4b** in 52% and 67% yields, respectively. Spirocyclic starting material **2c** resulted in the generation of **4c** in 61% yield. Benzaldehyde substituted product **4d** was obtained in 73% yield from bicyclo[4.2.0]octa-1,3,5-trien-7-ol derivative **2d**. Satisfyingly, 1,8-double-functionalized allenes **4e** and **4f** were successfully accessed, albeit in moderate yields. However, when the reaction was carried out with acyclic alcohol and cyclohexanol derivatives, no desired products were generated under the standard conditions.

Gram-scale preparation of **3a** was realized without a loss of efficiency (Scheme 4A). Diverse transformations of the product were realized, as exemplified in Scheme 4B. Alcohol derivative **5a** was generated via a reduction reaction with $NaBH_4$. Reductive amination yielded compound **5b**. A nucleophilic addition of methylmagnesium bromide to the aldehyde group generated **5c** in 95% yield. Epoxidation of **3a** with 1-benzyltetrahydro-1*H*-thiophen-1-ium bromide offered

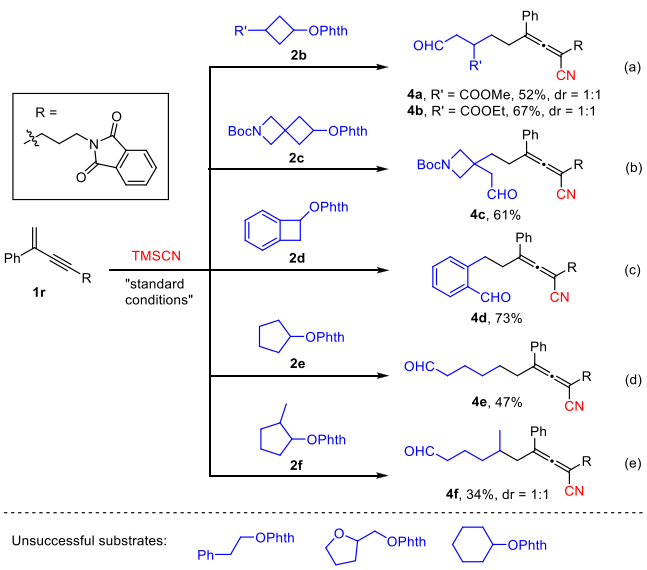
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Scheme 2. Scope of 1,3-Enynes[a]



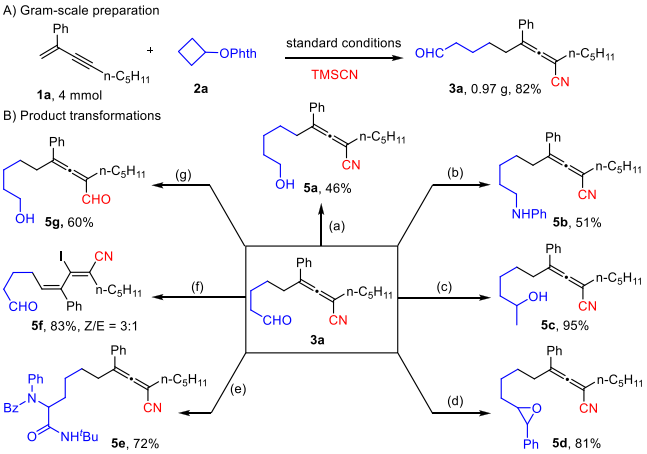
[a]Reaction conditions: **1** (0.1 mmol), **2a** (0.15 mmol), TMSCN (0.25 mmol), $Ir(ppy)_3$ (1 mol %), $Cu(MeCN)_4PF_6$ (10 mol %), L1 (12 mol %), DMF (2.0 mL), 20 W blue LEDs, N_2, 24 h. Yield of isolated products.

Scheme 3. Scope of Cyclic Alcohol Derivatives[a]



[a]Reaction conditions: **1r** (0.1 mmol), **2** (0.15 mmol), TMSCN (0.25 mmol), $Ir(ppy)_3$ (1 mol %), $Cu(MeCN)_4PF_6$ (10 mol %), L1 (12 mol %), DMF (2.0 mL), 20 W blue LEDs, N_2, 24 h. Yield of isolated products.

Scheme 4. Gram-Scale Preparation and Applications as Platform Molecules[a]

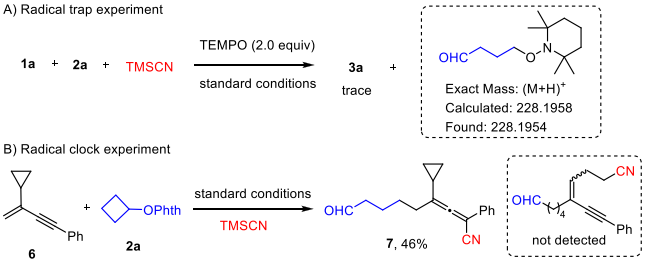


[a]Reaction conditions: (A) Gram-scale synthesis: **1a** (4 mmol), **2a** (6 mmol), TMSCN (10 mmol), $Ir(ppy)_3$ (1 mol %), $Cu(MeCN)_4PF_6$ (10 mol %), L1 (12 mol %), DMF (80 mL), 20 W blue LEDs, N_2, 24 h. (B) Transformations of **3a** (0.2 mmol scale): (a) $NaBH_4$ (2.0 equiv), MeOH (2.0 mL), 0 °C. (b) $PhNH_2$ (1.1 equiv), Na_2SO_4 (1 equiv), dichloromethane (0.4 mL), 2 h; then $NaBH_4$ (0.5 equiv), MeOH (0.4 mL). (c) MeMgBr (1.2 equiv), THF (2.0 mL), rt, 6 h. (d) 1-Benzyltetrahydro-1*H*-thiophen-1-ium bromide (1.5 equiv), 1,5,7-triazabicyclo[4.4.0]dec-1-ene (TBD) (1.5 equiv), dichloromethane (3.0 mL), rt, 12 h. (e) PhCOOH (1.0 equiv), $PhNH_2$ (1.0 equiv), *t*-butyl isocyanide (1.0 equiv), CH_3OH (0.4 mL), rt, 12 h. (f) NIS (1.3 equiv), dichloromethane (2.0 mL), 30 °C, 22 h, N_2; (g) DIBAL-H (2.5 equiv), dichloromethane (1.5 mL), −78 °C, 2 h, N_2.

compound **5d** in 81% yield. A chain-prolonging product **5e** was obtained in 72% yield, through an efficient four-component Ugi-reaction of **3a** with benzoic acid, aniline, and *t*-butyl isocyanide. Additionally, **3a** could be readily converted into 3-iodoalka-2,4-dienyl nitrile **5f** in 83% yield (Z/E = 3:1). The cyano group could be transformed to aldehyde with 2.5 equiv of DIBAL-H, along with the reduction of the alkyl aldehyde to a hydroxyl group, giving **5g** in 60% yield.

In order to gain further insight into the mechanism of this novel transformation, some control experiments have been conducted (Scheme 5). The allenation process was inhibited

Scheme 5. Mechanistic Experiments



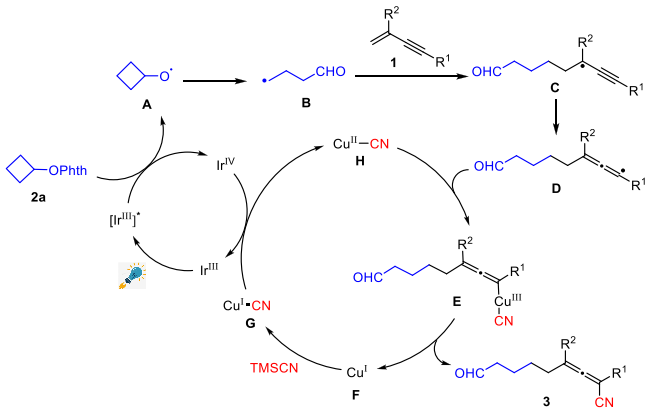
when the reaction was run in the presence of 2.0 equiv of the radical trapper TEMPO, and the TEMPO-trapped product was detected by HRMS analysis. Then, a radical clock experiment using cyclopropyl 1,3-enyne **6** was carried out. Cyclopropyl-substituted allene **7** instead of the ring-opening product was isolated in 46% yield, indicating that the propargyl−allenyl isomerization is faster than the ring-opening of the cyclopropyl group, which is crucial to the success of such an allene

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synthesis. Stern–Volmer fluorescence quenching experiments revealed that the excited state of $[Ir(ppy)_3]^*$ was effectively quenched by **2a** (for details, see Figure S3).

Based on the experimental results and literature reports,[19,20f] a possible dual catalytic mechanism was depicted in Scheme 6.

Scheme 6. Proposed Mechanism



The reaction was started by light-induced N–O bond cleavage, generating oxygen radical **A**. Then, a *β*-scission formed the aldehyde-functionalized primary radical **B**, which was trapped by 1,3-enyne **1** to give intermediate **C**. The propargyl–allenyl isomerization process furnished allenyl radical **D**, which was captured by Cu^{II} species to form allenyl-Cu^{III} species **E**. Subsequent reductive elimination generated the desired product **3** and regenerated the Cu^{I} species **F**. In the presence of TMSCN, intermediate **G** was generated, which was oxidized to Cu^{II} via a single electron transfer process.

In conclusion, we have developed efficient access to multiple functionalized allenes through a three component 1,4-dicarbofunctionalization of 1,3-enynes. By combining light-induced *β*-scission of cyclic butanol and pentanol derivatives with copper catalyst, 1,7- and 1,8-double-functionalized allenes were synthesized in the presence of TMSCN under mild reaction conditions with broad functional group tolerance. Late-stage modification of natural molecules and drugs was realized by using this method.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.3c04360.

Experimental procedures and characterization data (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Jian Wang − *Key Laboratory of Functional Molecular Solids, Ministry of Education, Anhui Laboratory of Molecule-Based Materials (State Key Laboratory Cultivation Base), College of Chemistry and Materials Science, Anhui Normal University, Wuhu 241002, P.R. China;* orcid.org/0000-0003-0806-9590; Email: wang_jian989@163.com

Yongjia Shang − *Key Laboratory of Functional Molecular Solids, Ministry of Education, Anhui Laboratory of Molecule-Based Materials (State Key Laboratory Cultivation Base), College of Chemistry and Materials Science, Anhui Normal University, Wuhu 241002, P.R. China;* orcid.org/0000-0001-9873-9150; Email: shyj@mail.ahnu.edu.cn

Authors

Yue Pu − *Key Laboratory of Functional Molecular Solids, Ministry of Education, Anhui Laboratory of Molecule-Based Materials (State Key Laboratory Cultivation Base), College of Chemistry and Materials Science, Anhui Normal University, Wuhu 241002, P.R. China*

Shumin Ding − *Key Laboratory of Functional Molecular Solids, Ministry of Education, Anhui Laboratory of Molecule-Based Materials (State Key Laboratory Cultivation Base), College of Chemistry and Materials Science, Anhui Normal University, Wuhu 241002, P.R. China*

Haixia Zhao − *Key Laboratory of Functional Molecular Solids, Ministry of Education, Anhui Laboratory of Molecule-Based Materials (State Key Laboratory Cultivation Base), College of Chemistry and Materials Science, Anhui Normal University, Wuhu 241002, P.R. China*

Qiaoli Xue − *Key Laboratory of Functional Molecular Solids, Ministry of Education, Anhui Laboratory of Molecule-Based Materials (State Key Laboratory Cultivation Base), College of Chemistry and Materials Science, Anhui Normal University, Wuhu 241002, P.R. China*

Heng Zhang − *Key Laboratory of Functional Molecular Solids, Ministry of Education, Anhui Laboratory of Molecule-Based Materials (State Key Laboratory Cultivation Base), College of Chemistry and Materials Science, Anhui Normal University, Wuhu 241002, P.R. China*

Xiaotian Xie − *Key Laboratory of Functional Molecular Solids, Ministry of Education, Anhui Laboratory of Molecule-Based Materials (State Key Laboratory Cultivation Base), College of Chemistry and Materials Science, Anhui Normal University, Wuhu 241002, P.R. China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.orglett.3c04360

Author Contributions

[‖]Y.P. and S.D. contributed equally to this work.

Notes

The authors declare no competing financial interest.

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